# Hossein Khoshhal

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## EDUCATION

**Carnegie Mellon University** **May 2025**
*Master of Science Data Analytics - Public Policy Management* *Pittsburgh, PA*
Coursework: Database Management, Econometrics, Operationalizing AI, Python, Management Consulting, Machine Learning, Optimization, Decision Analytics, Cloud Security, Product Management, Accounting & Financial Analytics, Agent Based Modeling

**George Mason University** **May 2022**
*Bachelor of Arts Philosophy, Bachelor of Arts International Politics - Minor: Data Analytics* *Fairfax, VA*

## CERTIFICATIONS

- Databricks Machine Learning Engineer Associate, AWS Certified Cloud Practitioner, Lean Six Sigma Yellow Belt

## SKILLS

- **Core Tools:** Python, SQL, PostgreSQL, Tableau, Databricks, AWS, ArcGIS, Docker, MongoDB, R, Gurobi, Excel, Git

## RELEVANT WORK EXPERIENCE

**Guidehouse – State & Local Government** **July 2025 – Present**
*Consultant* *New York, NY*
- Led process discovery and operating model design efforts for California's Integrated Claims Management System modernization, mapping 400+ user stories and developing future-state workflows across five statewide benefit programs.
- Served as the primary client-facing facilitator for the modernization effort, partnering daily with 30+ program directors, managers, analysts, and executive stakeholders to translate siloed processes, pain points, and business requirements into a unified future-state operating model.
- Assessed data governance controls, transaction processing logic, and data lake reporting pipelines supporting the Metropolitan Transportation Authority's congestion pricing program, tracing end-to-end transaction flows to identify control deficiencies, processing defects, and revenue leakage across one of the nation's largest tolling ecosystems.
- Identified approximately $16 million in uncollected toll revenue by uncovering failures in transaction processing logic, business rule enforcement, and exception-handling controls.
- Developed a Python-based procurement intelligence platform using Playwright, BeautifulSoup, and OpenPyXL that automated government solicitation monitoring, reducing opportunity scanning review time to under a minute, saving roughly 200 hours annually.
- Authored proposal approaches, staffing models, qualifications packages, and executive presentation materials supporting public-sector pursuits across Maryland, New York State, and New York City, contributing to three contract awards totaling approximately $3 million.

**FLM Haiti** **Jan. 2025 – July 2025**
*Systems Project Manager* *Pittsburgh, PA*
- Led design and deployment of an offline-capable electronic medical record platform for a rural healthcare clinic serving communities with unreliable power and internet connectivity.
- Architected a resilient local-cloud infrastructure using Flutter, PostgreSQL, Docker, and Supabase to ensure continuity of clinical operations during outages.
- Directed a multidisciplinary team of developers and data scientists through system design, implementation, stakeholder engagement, and deployment.

**National Institute of Standards and Technology (NIST)** **Aug. 2024 – Dec. 2024**
*Generative AI Security Analyst (ARIA)* *Remote*
- Conducted adversarial testing of frontier large language models including GPT-4o, Claude, Gemini, and Llama to identify vulnerabilities related to misinformation, policy circumvention, and AI safety controls.
- Developed attack methodologies and documented findings used to evaluate model robustness, security safeguards, and responsible AI deployment risks.

**Allegheny County Department of Human Services (DHS)** **Dec. 2023 – May 2024**
*Data Analytics Consultant* *Pittsburgh, PA*
- Leveraged Python, GIS, and prescription datasets to analyze substance-use disorder indicators across Allegheny County, developing analytical workflows that integrated geospatial and statistical analysis to identify providers with atypically high volumes of Schedule II narcotics prescriptions.
- Presented findings and data-driven recommendations to county stakeholders, translating complex analytical results into actionable insights that informed public health oversight, investigative prioritization, and intervention planning.

## SELECTED ANALYTICS RESEARCH

***AMIA 2025 ACCEPTED POSTER – RESILIENT HEALTH INFORMATION SYSTEMS FOR UNDERSERVED COMMUNITIES***
Research accepted for presentation at the 2025 American Medical Informatics Association (AMIA) Annual Symposium. Developed and evaluated a resilient health information system architecture for resource-constrained communities, demonstrating approaches to maintain data integrity, synchronization, and continuity of care despite unreliable internet connectivity and power infrastructure.

***CITY OF CHICAGO – OPTIMIZING HEALTHCARE ACCESS THROUGH DECISION ANALYTICS***
Developed a Gurobi-based Maximal Covering Location Problem (MCLP) model to identify optimal pharmacy locations in underserved communities. Leveraged GIS, OpenStreetMap, Census, and CDC datasets to evaluate accessibility, demographic need, and disease burden, improving equitable access to healthcare resources. Supported analysis with research on scalable urban planning strategies.